Paper ID: 1571326099

Title: Simulation-Driven Design and Functional Assessment of a Gait Simulator for Transtibial Prosthesis Evaluation.

Track: Human Motion and Rehabilitation Engineering

Dear reviewer,
We greatly appreciate your comments and constructive suggestions on our review article. We value your time and effort in providing us with these recommendations to improve the quality and comprehensiveness of our work. Below, we provide answers and clarifications to each of the points mentioned:

Reviewer 1

Comment 1: Introduction

I don't agree that gait asymmetries is necessarily bad for the person. Any citation for these? Since gait asymmetries are also found in healthy subject.

Other than that, the introduction seems ok. The author discuss from broad perspective of human locomotion until the problem in trans tibial prosthesis that requires gait simulator. However, the author needs to discuss the limitation of previously designed gait simulator. Why the [20] is complex?

Author's response:

We accept the point on gait asymmetry. It is not detrimental in itself and it does occur in able-bodied walking, and what we meant is that amputee gait amplifies it beyond that natural range. The original sentence merged both ideas into one and left the able-bodied range neither stated nor supported. We have separated them and added a citation of symmetry and limb dominance in able-bodied gait as +support.

The cite numbered [20] in the original manuscript, now [21], we did not mean to describe it as complex. We cited it for the trade-off it documents between the number of actuated degrees of freedom and system cost, and our phrasing was ambiguous. The direct comparison required is in Comment 4.

We also accept that our explanation of the limitations of previous simulators was too generic, since it listed possible shortcomings without saying which approach has which. We now state them per approach, all taken from references the paper already cited, so no new reference was needed here.

Author's action:

1. Asymmetry sentence rewritten; citations redistributed so that [1]–[3] support the general gait statement and [5], [6] the amputation-specific consequences

Original sentence:
“"In individuals with lower-limb amputation, these interactions are altered by the prosthetic device, often leading to gait asymmetries, increased energy expenditure, and abnormal loading patterns that may compromise mobility and long-term musculoskeletal health ~~[1]–[5]~~”

Revised:
“In individuals with lower-limb amputation, these interactions are altered by the prosthetic device. Although a degree of gait asymmetry reflects natural functional differences between the lower limbs in able-bodied walking [4], amputee gait is characterized by asymmetries amplified beyond that range, together with increased energy expenditure and abnormal loading patterns that may compromise mobility and long-term musculoskeletal health [5], [6].”

2. One reference added as [4]

   [4] H. Sadeghi, P. Allard, F. Prince, and H. Labelle, "Symmetry and limb dominance in able-bodied gait: a review," Gait & Posture, vol. 12, no. 1, pp. 34–45, Sep. 2000, doi: 10.1016/S0966-6362(00)00070-9.

3. Generic limitations sentence replaced by concrete limitations stated per approach, using references the paper already cited.

   Original sentence:
   “However, each approach presents ~~limitations regarding their biomechanical realism, implementation complexity, computational requirements, or experimental reproducibility [8]–[14]~~”

   Revised:
   “However, each approach presents specific limitations. Hardware-in-the-loop platforms require real-time coupling between the physical prosthesis and a numerical model [9]. Robotic gait simulators reproduce multi-planar motion, but at a capital cost that constrains their accessibility [10], [11]. Simulation-based approaches avoid hardware constraints altogether but do not evaluate physical prototypes [12]–[15]. Prosthesis simulators worn by participants without amputation introduce confounding factors that are difficult to separate from the device under test: leg-length discrepancies above 2 cm alter gait kinematics, the folded sound limb introduces an inertial artefact that can exacerbate step-length asymmetry, and the added mass increases metabolic cost [7]. Despite these constraints, such… ”

4. The paragraph no longer exists as such: it has been reduced to its single non-redundant statement, which is now the last sentence of paragraph 2 in point 3. Nothing has been added. This was done to make room for point 3 within the page limit.

   Original sentence:
   “~~Biomechanical simulators provide an attractive alternative for evaluating prosthetic~~ alignment, stiffness, damping, control strategies, and gait mechanics under controlled laboratory conditions. They enable systematic assessment of prosthetic components while reducing reliance on early-stage human testing ~~and facilitating the analysis of kinematic and kinetic variables that are difficult to obtain in conventional clinical experiments~~ [8], [9], [15]–[19].”

Revised:

"Despite these constraints, such platforms enable the systematic assessment of prosthetic alignment, stiffness, damping, and control strategies while reducing reliance on early-stage human testing [9], [10], [16]–[20]."

5. One sentence deleted, because it repeated the closing sentence of the preceding paragraph, and two wordings shortened. Nothing was added. Also done to make room for Item 3.

   Original sentence:

   "This work presents the design, implementation, and functional assessment of a 3-degree-of-freedom (3-DOF) gait simulator for transtibial prosthesis testing. The simulator provides two translational axes (horizontal and vertical) and one rotational axis (sagittal). ~~The proposed platform reproduces the essential kinematic and kinetic characteristics required for transtibial prosthesis evaluation using a mechanically simplified three-degree-of-freedom architecture~~ developed through a simulation-driven design methodology integrating finite element analysis, analytical transmission sizing, and functional verification. This study ~~provides a comprehensive description of~~ the simulator's design and implementation processes, as well as its functional assessment using videogrammetry and ground reaction force (GRF) analysis."

   Revised:

   "This work presents the design, implementation, and functional assessment of a 3-degree-of-freedom (3-DOF) gait simulator for transtibial prosthesis testing, providing two translational axes (horizontal and vertical) and one rotational axis (sagittal). The architecture was developed through a simulation-driven design methodology integrating finite element analysis, analytical transmission sizing, and functional verification. This study describes the design and implementation processes, together with a functional assessment using videogrammetry and ground reaction force (GRF) analysis."

Comment 2: Method

Figure 1 should include the prosthesis. Its hard to understand, but easier to see the simulator at Figure 4. Therefore, I suggest to include the prosthesis in Figure 1 also.

Marker location should be shown in a Figure. Some calculation regarding the inclination angle modelling using the marker is necessary. Its hard to understand which inclination angle do you mean?

<u>Author's response:</u>

We accept both points.

We named the inclination angle throughout the paper but never defined it. It is the orientation of the segment perpendicular to the tibial axis, formed by markers M3 and M4, measured against the horizontal axis of the image, which is equivalent to the deviation of the tibial segment from the vertical. The same definition applies to the two markers spanning the simulator platform, which is perpendicular to the prosthesis axis, so the same geometric quantity is measured on the subject and on the simulator. Marker placement is now shown in Fig. 4(b).

PONTIFICIA UNIVERSIDAD CATÓLICA DEL PERÚ

<u>Author's action:</u>

1. Fig. 1. Replaced by a new CAD rendering that includes the mounted transtibial prosthesis, labelled "Transtibial Prosthesis". Caption extended to state that the prosthesis is now included. Nothing removed.

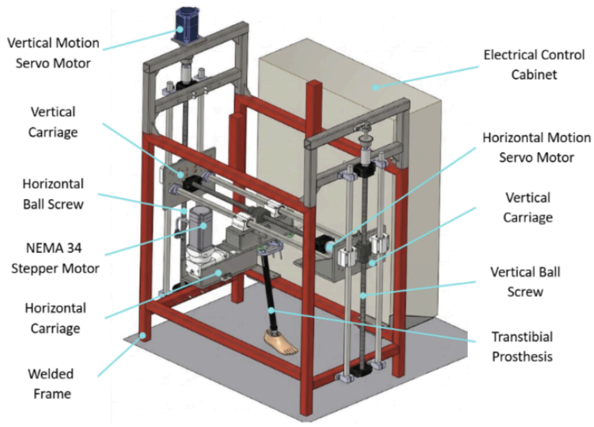
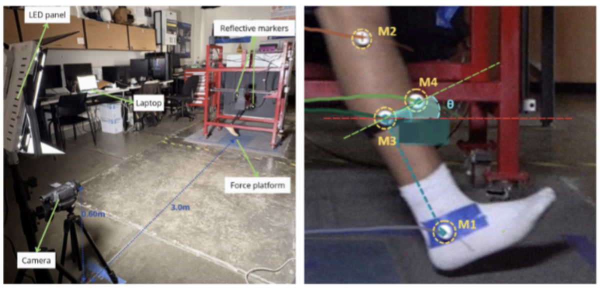


Fig. 1. CAD model of the final mechanical architecture and principal motion modules, including the transtibial prosthesis mounted on the platform.

2. Fig. 4. A second panel (b) added, with the placement of markers M1–M4 on the subject and the definition of θ; the caption extended to name both panels. Nothing removed.



(a) (b)

Fig. 4. (a) Experimental setup for kinematic and kinetic data acquisition. (b) Placement of markers M1–M4 on the reference subject and definition of the tibial-segment inclination angle θ.

3. Markers relabelled M1–M4 to match Fig. 4(b), and the operational definition of θ added, with the expression used to compute it, the sign convention, and the statement that the same definition applies to the simulator markers. What is removed is the anonymous designations and one sentence that named the computation without defining the angle.

Original sentence:

"Four reflective markers were placed on the reference subject: ~~the first~~ at the lateral malleolus and ~~the second~~ 42 cm proximal to it… ~~A third~~ marker was placed at the midpoint of the segment formed by these two markers, and a fourth marker was aligned with the third to form a segment perpendicular to the tibial segment, allowing the calculation of the inclination angle of the tibial segment. […] ~~The tibial-segment inclination angle was computed from the marker coordinates using the atan2 function, defined as positive above the horizontal reference and negative below it.~~"

Revised:

"Four reflective markers (M1–M4) were placed on the reference subject: M1 at the lateral malleolus and M2 42 cm proximal to it… M3 was placed at the midpoint of the segment formed by M1 and M2, and M4 was aligned with M3 to form a segment perpendicular to the tibial segment, directed anteriorly. On the simulator, two reflective markers were placed at the ends of the moving platform to track its trajectory under the same protocol. The tibial-segment inclination angle θ was defined as the orientation of the segment perpendicular to the tibial axis (M3–M4) relative to the horizontal image axis, $\theta = \mathrm{atan2}(y\_M4 - y\_M3, x\_M4 - x\_M3)$, positive above the horizontal and negative below; equivalently, the deviation of the tibial segment from the vertical. The same definition was applied to the two simulator markers spanning the moving platform, which is perpendicular to the prosthesis axis. Marker placement is shown in Fig. 4(b)."

Comment 3: Result

Figure 5: the Fz seems ok. The inclination angle also nice.

Author's response:

No changes have been requested. Figure 5 does look different in the original manuscript, as the second reviewer requested that improvements be made. The curves, as well as the statistics on which they are based, have not been changed.

Author's action:

Changes to Figure 5 are reported under Reviewer 2, Comment 7.

Comment 4: Discussion

Explain how the design is simpler, yet produced appropriate result? direct comparison with complex approach that you said in [20] is demanded.

Author's response:

The design is simpler in that frontal and transverse plane motion is not actuated at all. The result is still appropriate because the two assessed variables, the tibial inclination angle and the vertical ground reaction force, are governed by the three sagittal-plane coordinates that were retained; omitting the other planes removes nothing those measurements depend on.

For the direct comparison, the informative terms are the actuated coordinates and the target population, since those are the terms in which the two platforms can be set side by side. The three coordinates retained here, flexion–extension with vertical and horizontal translation, are the same ones identified in [21] as those required for prosthetic testing, applied there to prosthetic knees and here to transtibial prostheses. As noted under Comment 1, the [20] (now [21]) was cited for the trade-off it documents, not as an example of a complex approach.

Author's action:

1. The sentence claiming that the reduced-degree-of-freedom architecture reproduces the essential gait characteristics was deleted and replaced by an explicit statement of what is simplified, followed by the direct comparison with [21].

Original sentence:
"~~Furthermore, the results show that the proposed reduced degree of freedom architecture can reproduce the essential gait characteristics required for transtibial prosthesis evaluation.~~"

Revised:
"The kinematic fidelity reported above was obtained with an architecture simplified with respect to multi-degree-of-freedom robotic simulators: frontal- and transverse-plane motion is not actuated, while the three sagittal-plane coordinates governing the assessed variables are retained. These same three coordinates, flexion–extension with vertical and horizontal translation, have been identified as those required for prosthetic testing [21], in that case applied to prosthetic knees rather than to transtibial prostheses."

2. Four numerical values repeated verbatim from the Functional Assessment section were deleted, to make room for the comparison added in Item 1 within the page limit. Nothing was added, and the values remain reported in that section.

   Original sentence:
   "The functional assessment confirmed that the simulator accurately reproduces the reference subject's tibial-segment inclination angle during stance ~~(RMSE = 0.38°, r = 1.00)~~ and swing ~~(RMSE = 1.58°, r = 0.997), which were evaluated independently~~, with ~~inter repetition~~ ICC(3,1) values above 0.999 in both phases…"

   Revised:
   "The functional assessment confirmed that the simulator accurately reproduces the reference subject's tibial-segment inclination angle during stance and swing, with ICC(3,1) values above 0.999 in both phases…"

Comment 5: Conclusion
Is enough

Author's response:

No changes have been requested. However, the conclusion has been revised, as the second reviewer asked us to make some changes.

Author's action:

Changes to the Conclusion are reported under Reviewer 2, Comment 10.

PONTIFICIA UNIVERSIDAD CATÓLICA DEL PERÚ

Reviewer 2

Comment 1:

Please correct and complete Reference [2], which appears incomplete and may not represent the intended source.

Author's response:

What we had cited was not the book but a one-page review of it published in a journal (F. Ozyener, J. Sports Sci. Med., vol. 9, no. 2, p. 353, 2010), and the entry was also missing its authors. The intended source is the book itself, which is what supports the statement made in the Introduction.

Author's action:

1. The journal information is removed from the one-page review; the authors, the book's edition, and the publisher are added. The reference number remains the same.

   Original sentence:
   "[2] Gait Analysis: Normal and Pathological Function," ~~J. Sports Sci. Med., vol. 9, no. 2, p. 353, Jun. 2010~~"

   Revised:
   "[2] J. Perry and J. M. Burnfield, Gait Analysis: Normal and Pathological Function, 2nd ed. Thorofare, NJ, USA: SLACK, 2010."

Comment 2:

Use the terms accuracy, agreement, correlation, tracking error, and repeatability consistently and according to their statistical meanings.

Author's response:

We checked the five terms one by one, and two of them were being used incorrectly.

The first is agreement. It was applied to a comparison that is not one, since the simulator is commanded to follow a prescribed trajectory and is therefore not a second measurement method compared against a first on the same measurand. What we are quantifying is a tracking error, which is the term mentioned in the comment and the one we had omitted.

The second is that we reported an error metric without defining it. Those values are not an RMSE in degrees or in percentage of body weight, as had been assigned to them. They are the RMSE divided by the pointwise standard deviation of the reference, which makes the quantity dimensionless. The unit symbols were therefore wrong, though no numerical value changes.

The remaining three were used consistently. We use correlation only for the Pearson coefficient, which measures waveform similarity and not closeness, repeatability only for the ICC(3,1), on which we say more under Comment 9, and accuracy, as a descriptive adjective based on those two quantities, rather than as an independent term.

Author's action:

1. The sentence defining the metrics was reformulated: agreement removed, tracking error introduced, and RMSEnorm defined. The word trajectory, which appears earlier in the same sentence, was dropped at its second occurrence.

   Original sentence:

   "~~Agreement between the simulator output and the~~ reference ~~subject~~ trajectory was quantified using RMSE, the Pearson correlation coefficient (r), and the percentage of simulator data points within ±1 SD of the reference ~~trajectory~~"

   Revised:
   "The simulator's tracking error with respect to the reference trajectory was quantified using RMSE normalized by the pointwise reference standard deviation (RMSEnorm), waveform similarity using the Pearson correlation coefficient (r), and the percentage of simulator data points within ±1 SD of the reference, which expresses the same normalization point by point"

2. The unit symbols were removed from the five normalized RMSE values added. Nothing else changed in these sentences.The five numerical values are unchanged; only the unit symbols, which the metric does not carry, were removed.The intra-subject RMSE values reported for the reference subject (1.41° during stance, 2.53° during swing) are not normalized and remain expressed in degrees.

Where	Original	Revised
Abstract	RMSE values of 0.38~~º~~ and 1.58~~º~~	RMSEnorm values of 0.38 and 1.58
Abstract	an RMSE of 21.87~~%BW~~	an RMSEnorm of 21.87
Results	RMSE = 0.38~~º~~	RMSEnorm = 0.38
Results	RMSE = 1.58~~º~~	RMSEnorm = 1.58
Results	RMSE = 21.87~~%BW~~	RMSEnorm = 21.87

Comment 3:

Describe signal-processing procedures, including:
- o filtering;
- o cutoff frequency;
- o gait-event detection;
- o time normalization;
- o resampling;
- o treatment of missing marker data.

Au

We accept that we omitted this. The six items are now covered in the order requested, and two of them need a comment here.

The first is the cutoff frequency, which we give for the force signal but not for the kinematic filter. The marker coordinates were smoothed with a Savitzky–Golay filter, which is fully specified by its polynomial order and its window length, the two values we report, and for which a cutoff frequency is not a design parameter.

The second is gait-event detection, done with a threshold on the filtered vertical force. This is the kinetic criterion used as reference standard in [22], the only point in this comment requiring a source not already in the paper.

Author's action:

1. A signal-processing paragraph added. Nothing removed.
   Added text:
   "Marker coordinates were smoothed with a third-order Savitzky–Golay filter over a 9-frame window, and the vertical force with a zero-phase fourth-order Butterworth low-pass filter at 15 Hz. Initial contact and toe-off were detected on the filtered force using a 20 N threshold [22]. Stance and swing were time-normalized to 0–60 % and 60–100 % of the gait cycle and resampled by piecewise cubic interpolation. Marker occlusions were resolved by manual re-tracking in Kinovea."

   One reference added:
   " [22] N. Zahradka, K. Verma, A. Behboodi, B. Bodt, H. Wright, and S. C. K. Lee, "An evaluation of three kinematic methods for gait event detection compared to the kinetic-based 'gold standard'," Sensors, vol. 20, no. 18, Art. no. 5272, Sep. 2020, doi: 10.3390/s20185272."

Comment 4:

Please complete the information of walking speed, gait-cycle duration, stance duration, and simulator execution duration.

Author's response:

Added, although not exactly as requested, and we prefer to point out the difference rather than let it go unnoticed.

Every quantity is reported per phase, with no gait-cycle duration. Stance and swing were recorded in separate captures, and the simulator, likewise, executes them as two separate programmed sequences, rather than as a continuous motion. Since no continuous cycle was ever acquired, a single cycle duration would attribute to the data a continuity that neither the acquisition nor the device has.

Walking speed comes from those same captures, as the horizontal displacement of the tibial segment over the recorded cycles divided by the duration of the phase, so it is reported per phase as well. The two values differ appreciably because the tibia advances over the planted foot during stance and swings forward freely during swing; a single figure would hide that difference rather than describe it.

One clarification on the fourth quantity. Speed applies to the reference subject only, because the simulator does not locomote and reproduces the recorded trajectory in place, so its counterpart quantity is the execution duration and not a speed.

Author's action:

1. A sentence added with the requested quantities, reported per phase. Nothing removed.

   Added text:
   "Stance and swing were captured separately, lasting 0.95 s and 0.55 s in the subject and 28.8 s and 15.9 s in the simulator. The subject's mean walking speed, computed from the horizontal displacement of the tibial segment over each phase, was 0.48 m/s in stance and 0.97 m/s in swing."

Comment 5:

Explain the rationale for using the percentage of points within ±1 SD. This is not a standard measure of agreement and is based on only ten cycles from one participant.

Author's response:

The percentage is not a standard measure of agreement, nor is it intended to be one. It is not an independent measure at all, but the same normalization already in use, read point by point. The error metric divides the error by the pointwise standard deviation of the reference, and this percentage is the fraction of the cycle in which that error stays below one standard deviation. It describes how the error is distributed along the cycle rather than adding a criterion, and it is not offered in place of a proper method-comparison analysis.

On the second part, ten cycles from one participant cannot support inference beyond this dataset, and we draw none. The ±1 SD band describes this dataset alone, never as a validation criterion. The single participant is now stated explicitly in the Conclusion, as reported under Comment 10.

Author's action:

1. The rationale for the ±1 SD percentage added as a closing clause to the sentence already reformulated under Comment 2. Nothing removed for this comment; the word trajectory struck below was dropped under Comment 2.

   Original sentence:

   "… and the percentage of simulator data points within ±1 SD of the reference ~~trajectory~~."

   Revised:
   "...and the percentage of simulator data points within ±1 SD of the reference, which expresses the same normalization point by point"

Comment 6:

Remove or support the claim that the platform is cost-effective. A bill of materials or approximate total system cost would be useful.

Author's response:

Of the two options offered, we take the first and remove the claim. We have no verified bill of materials for the platform, and an estimated figure would not support the claim in the terms requested.

The claim appeared once, in the Introduction, and is replaced by reduced-degree-of-freedom, which describes what the architecture demonstrably is and is the term the Abstract already used.

Author's action:

1. The claim of cost-effectiveness removed and replaced by the reduced-degree-of-freedom description. This is the only occurrence of the claim in the paper.

   Original sentence:
   "…capable of reproducing the essential kinematic and kinetic demands of gait within a ~~reduced and cost-effective~~ mechanical architecture."

   Revised:
   "...capable of reproducing the essential kinematic and kinetic demands of gait within a reduced-degree-of-freedom mechanical architecture."

Comment 7:

Improve Figure 5 by adding:

- simulator variability bands;
- peak values;
- timing annotations;
- an error or residual curve.

Author's response:

We have added all four elements. The curves were recomputed with the same processing that produced the original and every statistic reported is identical.

The variability band of the simulator had been computed all along but never displayed, which left the device looking as though it had none. It now appears in all three panels, beside the reference band.

The maximum values are labeled on the two panels showing the extremes, each with the time at which the value occurs in parentheses, expressed as a percentage of the operating cycle, which is the timing annotation requested. Panel b labels the minimum inclination reached by each curve, which puts a number on the deviation attributed in the Discussion to the mechanical limit of the sagittal axis.

The residual curve is drawn as a strip below each panel, showing the pointwise difference between simulator and reference on a shared horizontal axis, so that where the error concentrates along the cycle can be read directly instead of inferred from two overlaid curves.

Panel C includes a second vertical scale in newtons, which is the one referred to in comment 8.

Author's action:

1. Figure 5 was revised based on the feedback received

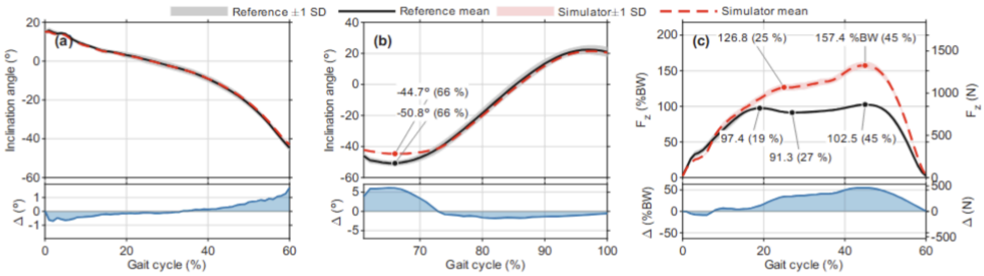


Fig. 5. Functional assessment of the gait simulator. Tibial-segment inclination angle during (a) stance and (b) swing, and (c) vertical ground reaction force during stance. Markers label each extremum, with its instant of occurrence in parentheses as a percentage of the gait cycle. Δ is the pointwise residual, simulator − reference.

Comment 8:

Report ground reaction force in both Newtons and percentage of body weight.

Author's response:

Force is now expressed in both units. The body weight used for normalization is 86 kg, or 843.7 N, and this is now specified in the manuscript, so any percentage we provide can be converted directly.

Panel c of Figure 5 includes a second vertical axis in newtons on its right side, so that each force in that panel can be read in either of the two units.

Author's action:

1. Section Functional Assessment, now gives the simulated body weight in kilograms and in Newtons, which is the value the %BW normalization is based on.
2. Panel c of Figure 5 includes a second vertical axis in Newtons, applied to the force curves and to their residual strip. Nothing was removed in either place. The forces quoted in the text keep their %BW values, with the Newton reading of each one given by that second axis.

Comment 9:

Use consistent terminology for the repeatability analysis, for example “intra-device inter-trial repeatability.”

Author's response:

We have adopted the suggested term. "Inter-repetition" does not identify the source of variation we quantify and could describe several different analyses. We evaluate a single device across repeated trials of the same programmed trajectory, so intra-device inter-trial repeatability is the accurate description. No numerical value changed and nothing was removed, since only the designation of the analysis was unified.

Author's action:

1. The term carries the new designation.

Original sentence:
"…with correlation coefficients of 1.00 and 0.997 and ~~inter-repetition~~ ICC(3,1) values above 0.999"

Revised:
"…with correlation coefficients of 1.00 and 0.997 and intra-device inter-trial ICC(3,1) values above 0.999."

2. Dropped without replacement, since the designation has already been given four lines above.

   Revised:
   "…a correlation coefficient of 0.9501, and an ~~inter-repetition~~ ICC(3,1) of 0.9984 were obtained."

3. The designation added at its first use in the body, which is the sentence that defines what the ICC measures. Nothing removed.

   Revised:
   "In addition, the intra-device inter-trial repeatability of the simulator across its ten programmed repetitions was quantified independently using the intraclass correlation coefficient ICC(3,1).."

4. The same deletion in all four, with ICC(3,1) standing alone once the designation has been defined. Nothing added.

   Revised:
   "The ~~inter-repetition~~ ICC(3,1) was 0.999, confirming high repeatability across the ten simulator trials." "…and 72.50 % of points within ±1 SD, with an ~~inter-repetition~~ ICC(3,1) of 0.999." "…with an ~~inter-repetition~~ ICC(3,1) of 0.9984, confirming consistent force output across the ten simulator repetitions." "…with ~~inter-repetition~~ ICC(3,1) values above 0.999 in both phases, confirming high repeatability of the simulator's motion across trials."

Comment 10:

Revise the conclusion to reflect the preliminary nature of the study.

An alternative more appropriate conclusion would be:
"The study demonstrates the preliminary feasibility of a repeatable, position-driven three-DOF gait simulator. Further independent validation of all motion axes, improved kinetic agreement, structural durability testing, and experiments comparing different prosthesis configurations are required before the platform can be considered validated for transtibial prosthesis evaluation."

<u>Author's response:</u>

We have adopted the proposed text almost verbatim, since our previous conclusion overstated the evidence provided by a study conducted with a single device and a single participant. There are three small departures from it, which we declare here. We added "assessed with a single participant", the limitation raised in Comment 5. We wrote "improved kinetic tracking" instead of "improved kinetic agreement", since agreement was removed from the manuscript in Comment 2 for the same reason.

And we used "3-DOF", the abbreviation already defined earlier in the Conclusion. The three future-work topics from the previous version are retained, condensed into one sentence.

Author's action:

1. The last sentence has been removed, as it repeated what had already been stated in the paragraph. Nothing has been added.

   Removed text:
   "~~The obtained RMSE and correlation coefficients confirmed the platform's fidelity, and the high inter-repetition ICC values demonstrated its repeatability.~~"

2. "Accurate reproduction" replaced by "low tracking error in the reproduction", following Comment 2.

   Original sentence:
   "The functional assessment demonstrated ~~accurate~~ reproduction of the tibial inclination angle during both the stance and swing phases."

   Revised:
   "The functional assessment demonstrated low tracking error in the reproduction of the tibial inclination angle during both the stance and swing phases."

3. The original statement has been replaced by the proposed text, and the section on future work has been retained in summary form.

   Original sentence:
   "~~These results indicate that the proposed simulator constitutes an accurate, repeatable, and controlled experimental platform for the engineering evaluation of transtibial prostheses prior to comprehensive biomechanical validation.~~ Future work will ~~focus on further assessing the platform using~~ multiple subjects with varying anthropometric characteristics ~~and extending it to reproduce~~ gait patterns associated with foot pathologies. In addition, the proposed platform will be used to evaluate prototype powered transtibial prostheses equipped with closed-loop control."

   Revised:
   "The study demonstrates the preliminary feasibility of a repeatable, position-driven 3-DOF gait simulator, assessed with a single participant. Further independent validation of all motion axes, improved kinetic tracking, structural durability testing, and experiments comparing different prosthesis configurations are required before the platform can be considered validated for transtibial prosthesis evaluation. Future work will also extend the assessment to multiple subjects with varying anthropometric characteristics, gait patterns associated with foot pathologies, and prototype powered transtibial prostheses equipped with closed-loop control."